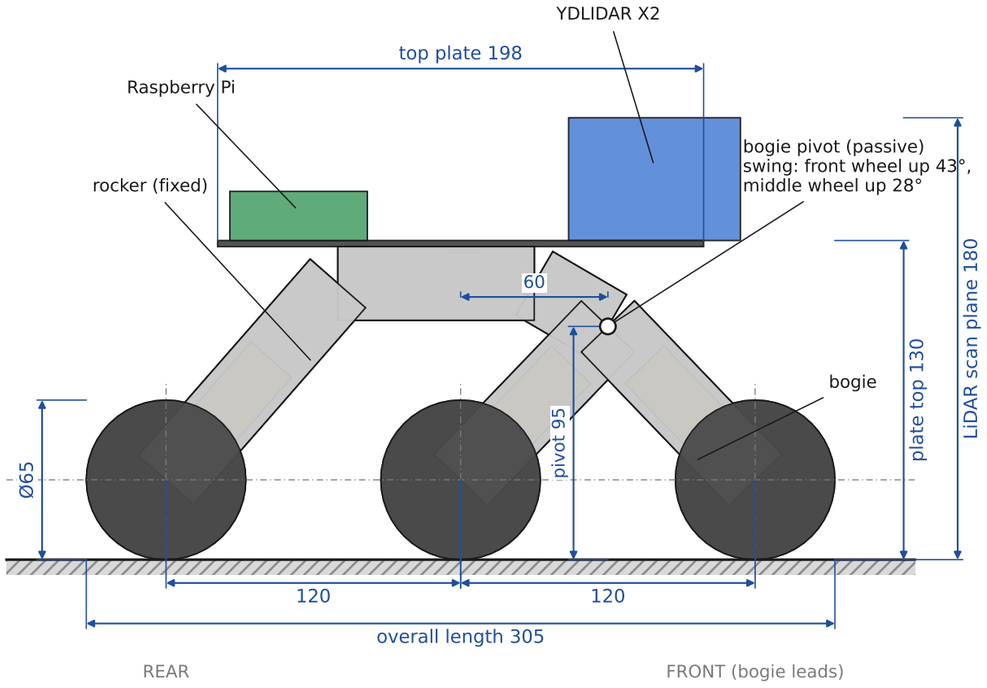
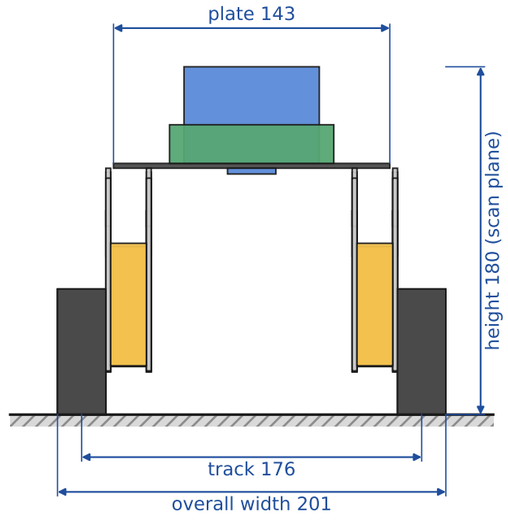
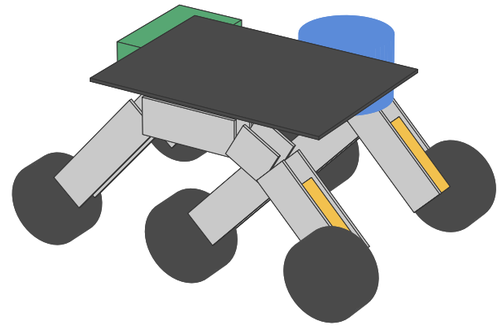
SIDE ELEVATION (from the right side; front →)



FRONT ELEVATION

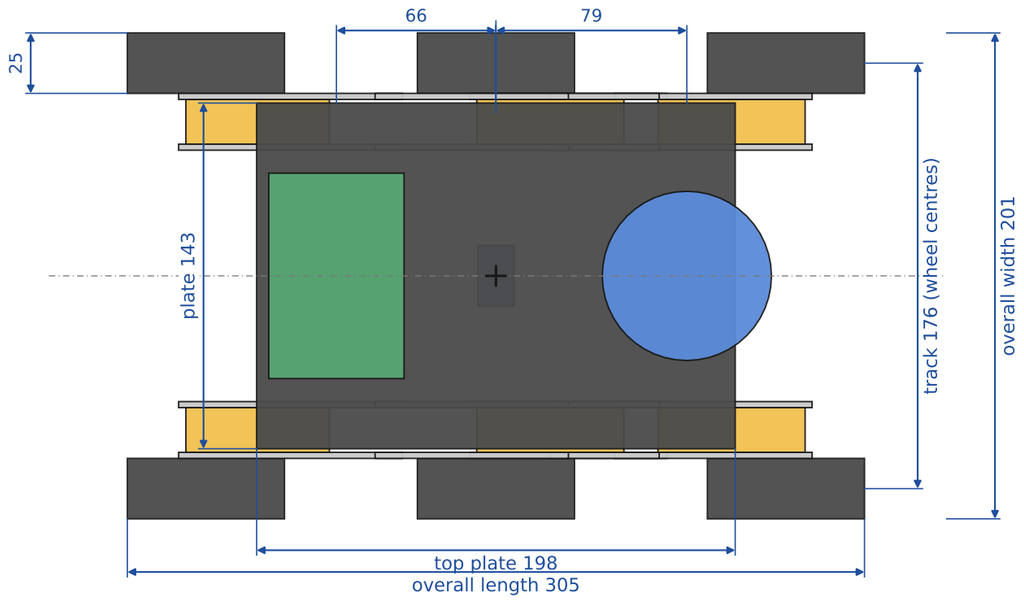


3-D VIEW



PLAN (top view; front →, robot left ↑)

+ robot centre (base_footprint); IMU (MPU-9250) under the plate here, chip facing down.
Green: Raspberry Pi. Blue: YDLIDAR X2. Yellow: BO motors between the side plates.



EXPLORER ROVER — GENERAL ARRANGEMENT

6-wheel skid-steer, rocker-bogie suspension (Kit4Curious-style), bogie end leads

Dimension	Value (mm)	Source
Overall length × width × height	305 × 201 × 180	derived
Top plate (acrylic)	198 × 143 × 2.5	measured
Plate top above ground	130	measured
Wheels (×6, BO geared motors)	Ø 65 × 25	measured
Axle spacing	120 + 120	measured
Track, wheel centres	176	measured
Bogie pivot	95 high, 60 ahead of middle axle	measured / derived
Bogie swing	43° front up / 28° middle up	from photos (±5°)
Side-plate bars	30 wide × 2.5 thick	measured
LiDAR (YDLIDAR X2)	Ø 70 × 50, 79 ahead of centre	assumed size
LiDAR scan plane	180 above ground	assumed height
Raspberry Pi	85 × 56 × 20, 66 behind centre	assumed
Motor bodies	65 × 18.5 × 22	assumed
Mass (estimate)	≈ 851 g	assumed

All dimensions in mm. Frames: x forward, y left, z up (REP-103); origin = base_footprint (ground, under the middle axle).
Part outlines are projections of the URDF visual geometry (explorer_description/urdf/explorer.urdf.xacro),
joints at 0. Sim-only: Gazebo DiffDrive effective track (skid_track) is a calibration value, not geometry.

Drawn: generated by rover_drawing.py | Scale: not to print scale, proportions exact | Units: mm